



ASME 2026 STUDENT HACKATHON · AUTODESK NEURALCAD-EDIT

UConn CAD PACK

A three agent loop that edits a solved B-rep from one line of customer intent, and measures its own work while it does it.

48 / 48 tasks scored

diff F1 **0.390**

best published model **0.184**

\$0.83 a task



RESULTS

48 of 48 tasks, scored on diff F1

0.390

our mean diff F1
all 48 tasks scored

0.184

best published model
GPT-5.2, the same 48 tasks

16

solved outright
diff F1 at or above 0.60

\$0.83

per task
\$39.98 for the whole benchmark

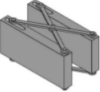
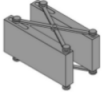
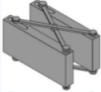
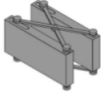
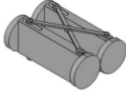
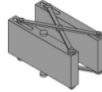
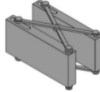
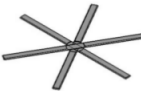
Second only to the second human expert at 0.595, which is the agreement ceiling for this metric rather than a competitor. Chamfer similarity 0.98 and volume F1 0.91 are healthy and are not the point: both sit near 1.0 for every method, including the ones that never edited the part at all.



THE FIVE TASKS THE ORGANISERS NAMED

All five, every method, with the score under each part

Our mean on these five is **0.473**, the best benchmark model reaches 0.181, and the second human 0.622.

	INPUT	GROUND TRUTH	OURS	GPT-5.2	GEMINI 3 PRO	CLAUDE SONNET 4.5	SECOND HUMAN
MEDIUM	"Please convert the round edges of the gear into straight spur gear teeth."						SUJ2G2 · 09987
			 0.131	 0.039	 0.000	 0.007	 0.870
MEDIUM	"Add cylindrical heads on the long pins to prevent link arms against slipping off."						3YH2WF · 73335
			 0.579	 0.864	 0.000	 0.000	 0.561
EASY	"Prolong black lever sticking out to the front by 5 cm for better manipulation."						B7A2N7 · 74133
			 0.797	 0.000	 0.000	no geometry returned 0.000	 1.000
HARD	"Add a third rotor blade with the same design as the other two, radii on all four long edges, and thin the central stack to..."						F332D3 · 07142
			 0.783	 0.000	 0.014	no geometry returned 0.000	 0.150
HARD	"Add a connecting hole of 1.7 mm diameter and apply 0.1 mm grooves to increase grip."						ZK22J6 · 74422
			 0.073	 0.002	 0.002	no geometry returned 0.000	 0.531



WHAT WE ADDED

Six components we added to the framework

Part inspector IT CAN POINT AT THINGS

Measures the solid into a text index of holes, faces and edges.

Example: the plan can say **chamfer edge 68, 119, 127** instead of “chamfer the holes”.

Strategist A PLAN, NOT A WISH

Turns one line into ordered CAD sub-goals, each with tags and an envelope.

Example: **“add a flange”** becomes cut the pocket, then add the flange, then drill the four holes.

Three agents, one job FAILURES STAY SEPARABLE

One plans, one writes CadQuery for a single sub-goal, one judges.

Example: the executor **never sees the rest of the plan**, so a bad plan cannot hide behind a good build.

Syntax and API check BEFORE ANYTHING RUNS

Reads the CadQuery the model wrote and blocks the errors that never execute.

Example: an unbalanced `.faces()` chain, or a method that does not exist in this CadQuery build, is caught **without spending a run**.

QA inspector IT SAYS WHAT IS WRONG

An independent call with no stake in the edit passing, judging pictures and numbers.

Example: **“the chamfer landed on the outer rim, not the hole mouth”** goes back as the next instruction.

A skill set, picked by THE RIGHT RECIPES, NOT ALL OF THEM

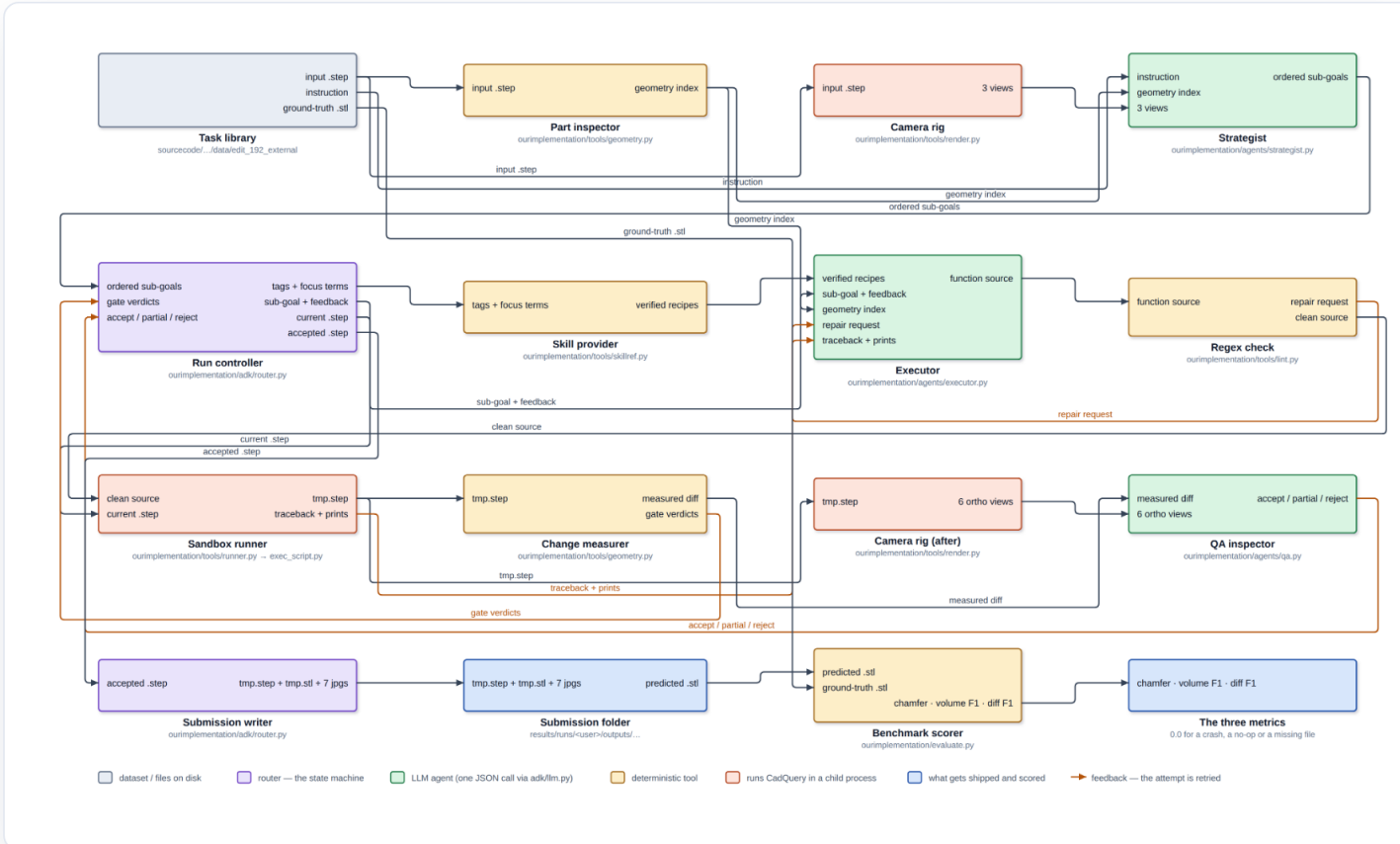
Every sub-goal the strategist writes carries tags. Those tags pull only the verified CadQuery recipes that fit it, inside a token budget.

Example: a sub-goal tagged **cut-hole-slot** is handed the hole cutting recipes only, and never the fillet cookbook.



THE FRAMEWORK, END TO END

Sixteen blocks, every wire labelled, and what holds it together



The models we run

Strategist gpt-5.6-sol

Executor gpt-5.2

QA gpt-5.2

The published baseline runs that same gpt-5.2.

Underneath, costing nothing

Six deterministic tools and six gates. A gate rejects an attempt **without a model call**.

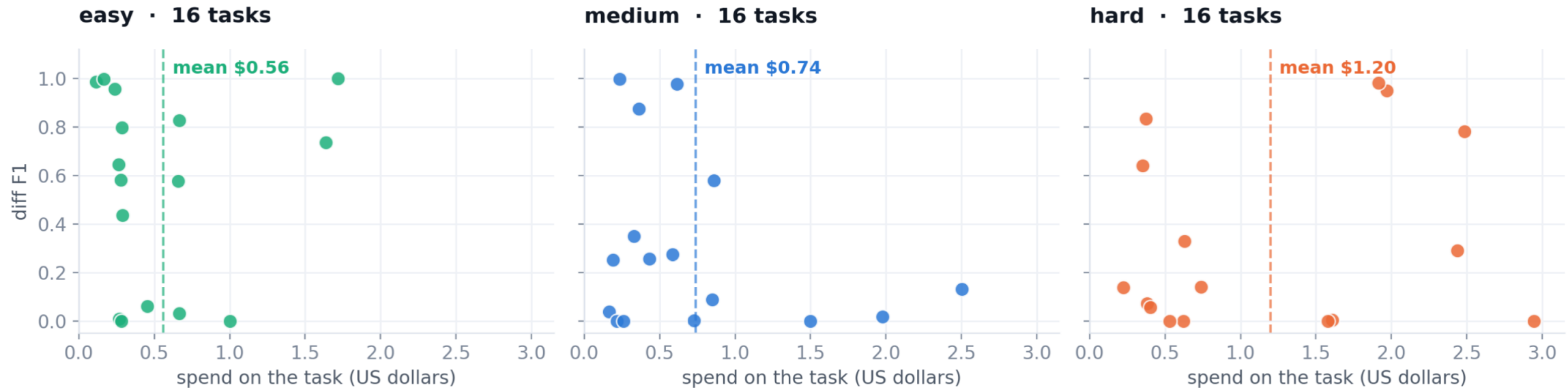
What bounds the spend

5 sub-goals · 3 attempts each · 2 replans · 4,100 token recipe budget



WHAT IT COSTS, AND WHAT HOLDS IT THERE

Cost efficiency is a design constraint, not an outcome



\$0.83

per task

\$39.98 for the whole benchmark

6

gates that cost nothing

a bad attempt is rejected without a model call

3

attempts per sub-goal

then the router rolls back to the best one

4,100

token recipe budget

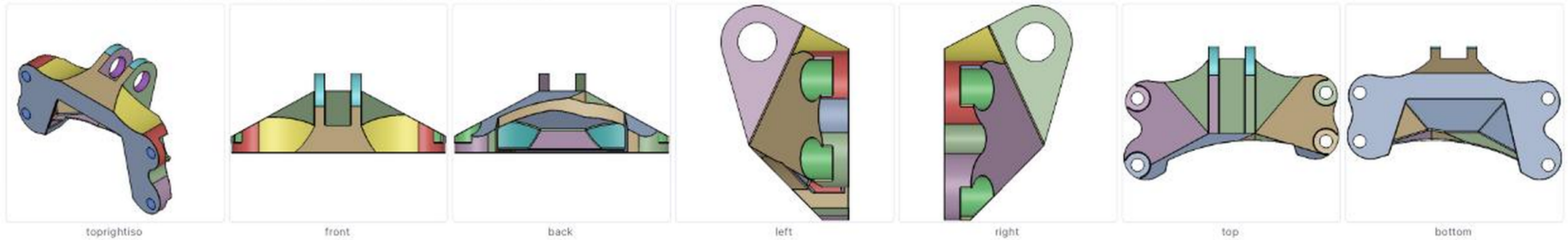
skills are selected, not pasted in whole

Every task priced from its own token counts at the rates in config.py.



ONE REQUEST

The part, the plan it became, and the score



What the strategist actually plans from: every feature family in its own colour, seven views, measured rather than guessed.

The whole input

“Add 0.2 mm chamfer to the hole edges to improve fitting.”

One line of text and a solved B-rep. No feature tree, no sketches, nothing named.

The plan, split into sub-goals

1. Apply a 0.2 mm 45° chamfer to all listed hole-mouth circular edges: edge_idx [68, 119, 127, 149, 152, 254, 256, 300, 120, 129, 285, 287, ...], 36 in all. Keep the chamfer on the hole entry and exit rims, not on unrelated blend or fillet artefact edges. [tags: [fillet-chamfer](#) · [done](#), 1 attempt]

0.646

diff F1

the metric that reads
edits

0.989

chamfer sim

near 1.0 for everyone

0.998

volume F1

near 1.0 for everyone

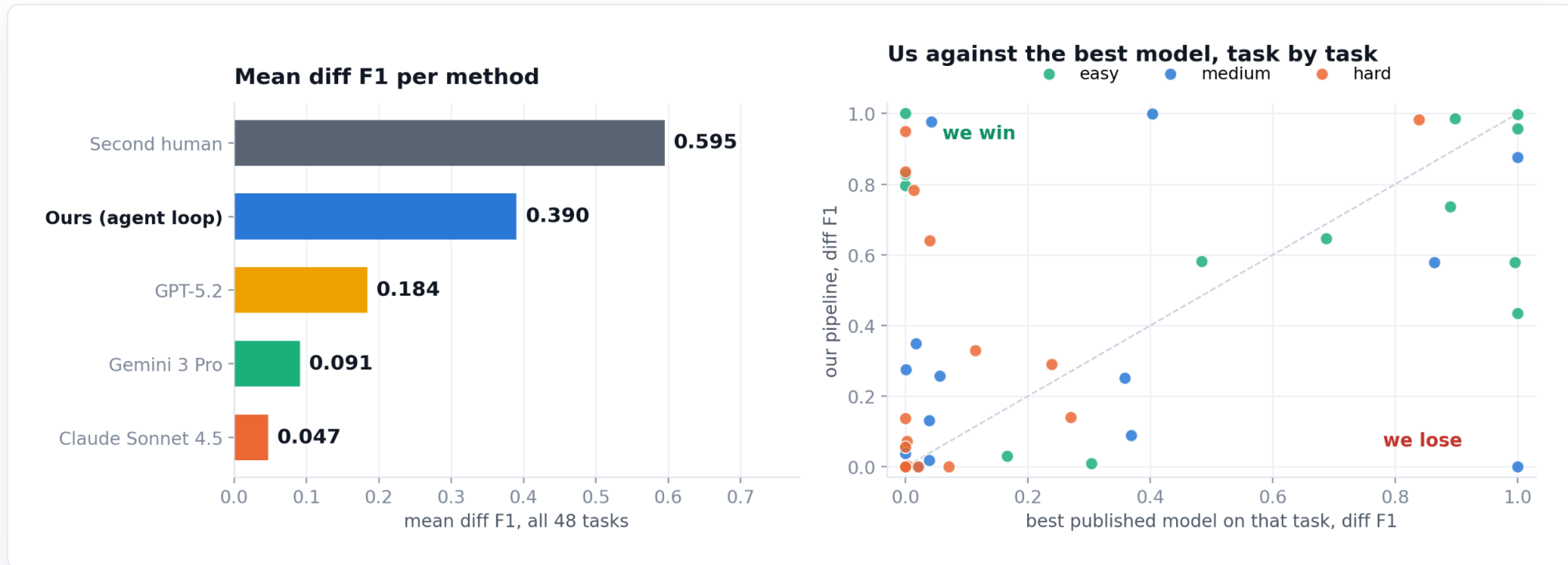
What it cost

Three model calls, 44,953 tokens, **\$0.18** for this run, start to finish.



THE WHOLE BENCHMARK

Ranked on diff F1, and task by task against the best model

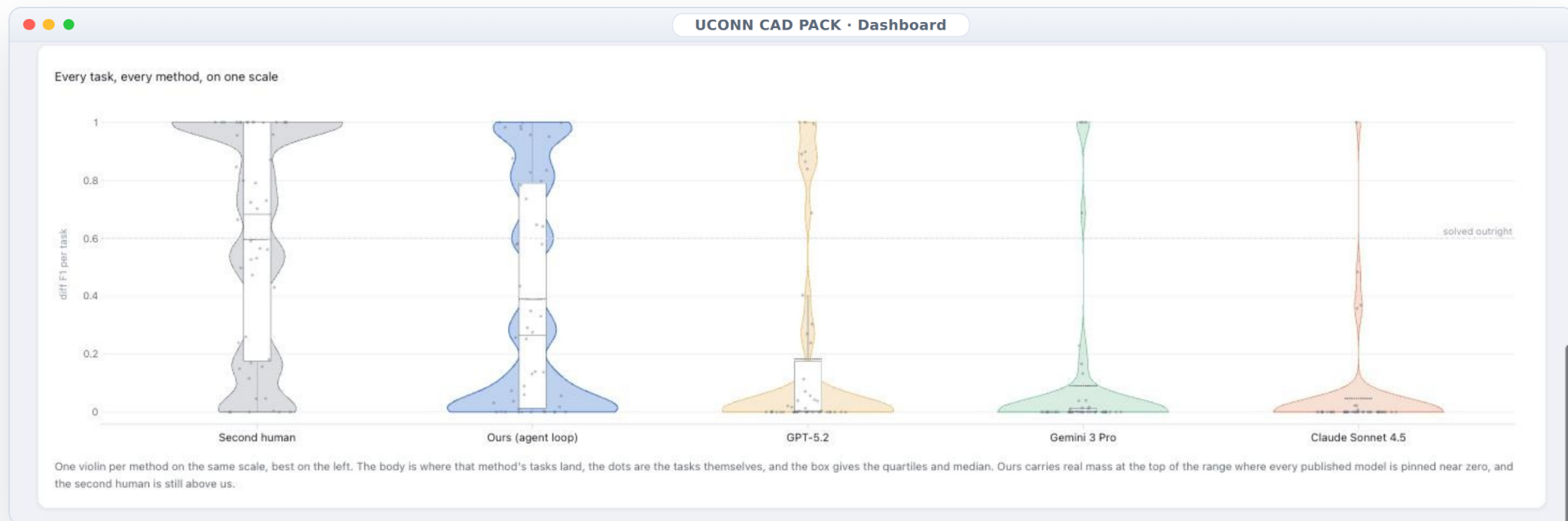


Left, mean diff F1 per method. Right, every task against the best published model on that same task: above the diagonal is ours. Drawn from the same data the dashboard plots, at print resolution.



DISTRIBUTION

A mean can hide everything, so here is every task



One violin per method on the same scale, the dots are the tasks themselves, the box gives the quartiles and median. Captured live from the UCONN CAD PACK dashboard, all 48 tasks scored.



BY DIFFICULTY

The harder the edit, the wider the margin

DIFFICULTY	OURS	GPT-5.2	SECOND HUMAN
easy · 16 tasks	0.540	0.299	0.867
medium · 16 tasks	0.302	0.155	0.422
hard · 16 tasks	0.326	0.097	0.497

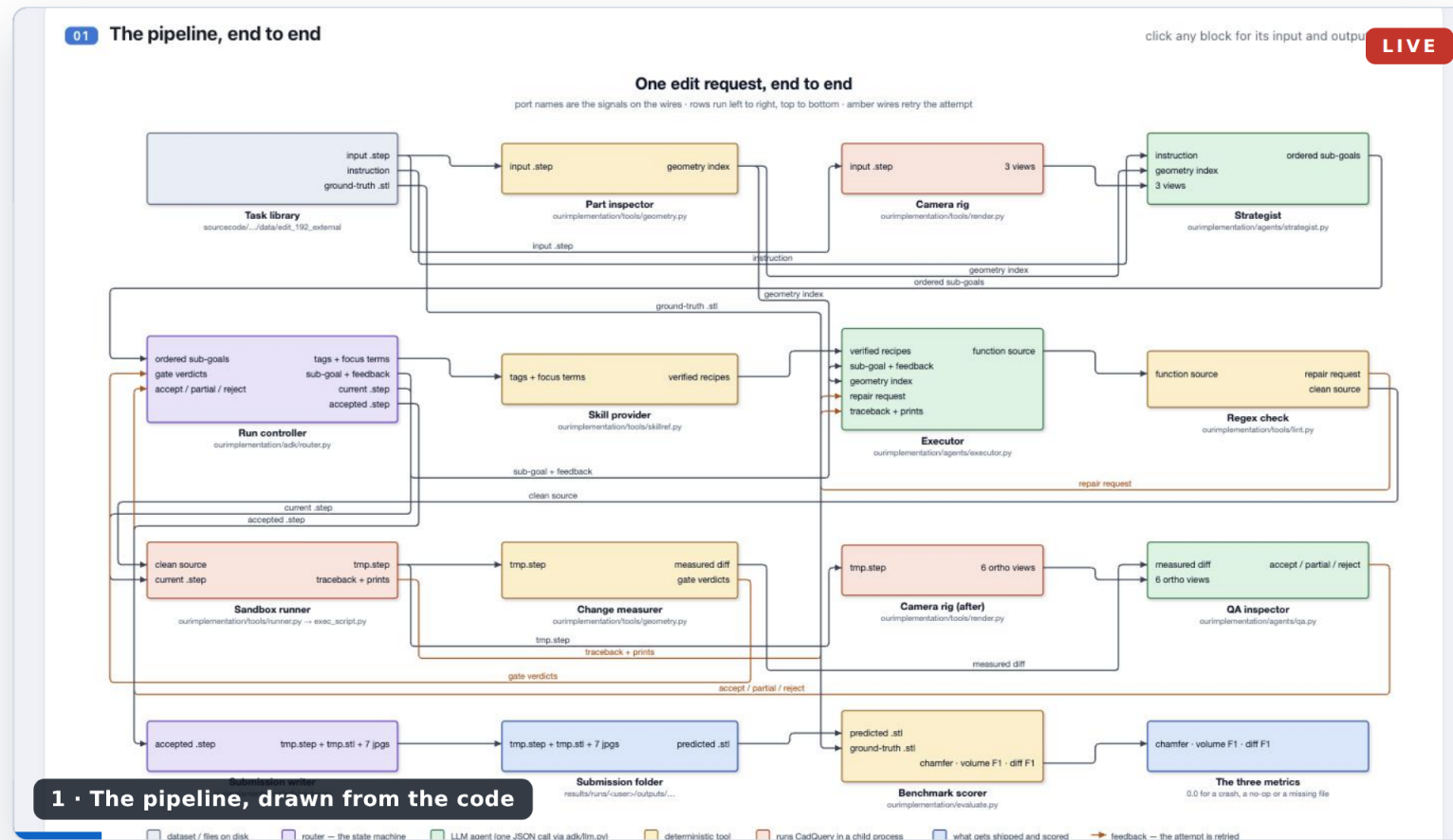
- Our hard band scores **above** our medium band, which is the loop doing the work rather than the model
- On hard edits we hold **0.326** while the published baseline falls to **0.097**
- 5 of the 48 are degenerate: a second human also scores 0.0 on them, so no pipeline change moves them
- On a separate task the ground truth diff is zero voxels, so **changing nothing scores 1.0** and any real edit scores zero



DEMO

45 seconds inside the dashboard

Every tab, one feature each. It runs the real pipeline, not a mock.





GIVEN MORE TIME

What we would test next, in order

Fine tune the strategist

PLANNING IS WHERE THE
LOSS IS

QA checks the build against the sub-goal, so a faithful build of a wrong plan is invisible to it. We found Fusion derived CAD datasets that clean up into plan supervision.

A mixture of experts in the executor

MORE CANDIDATES TO
CHOOSE FROM

A second vision capable model generating alongside gpt-5.2, then choosing between them. The only blocker is that every role shares one endpoint today, and a routing proxy fixes that.

Finish the selector

ALREADY PROTOTYPED

Pick the candidate that agrees most with the others. The kernel is already verified against the benchmark's own metric, and it needs at least three distinct candidates, which is exactly what a second expert provides.



Thank you

Everything on these slides is live and open: the dashboard, the handoff pack, the code and the scores.

PROJECT SITE

kiarash99naghavi.github.io/UConn-CAD-PACK



Scan for the dashboard, the handoff pack and the code